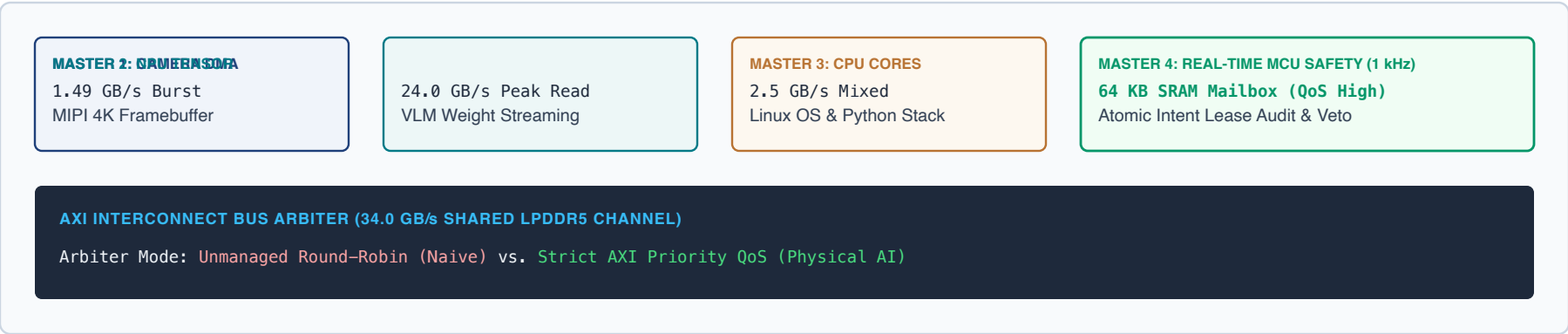


UNIFIED MEMORY ARCHITECTURE (UMA) BUS CONTENTION & AXI QoS ARBITRATION

Guaranteed < 2 μ s preemption for real-time safety mailboxes under heavy multi-master memory traffic



✗ UNMANAGED FIFO: MEMORY STARVATION & JITTER

DRAM Queue: [Camera DMA] [Camera DMA] [NPU] ✗ [MCU Blocked]
Camera bursts lock out MCU safety access for up to 4.8 ms

- **Priority Inversion:** 1 kHz safety tick stalled by uncompressed video.
- **Inference Tail Spikes:** NPU weight fetch stalls \$implies\$ P99 explodes to 135 ms.
- **Physical Breach:** Exceeds dynamic stability deadline \$implies\$ Hardware E-Stop.

✓ STRICT AXI QoS: DEDICATED BANDWIDTH & < 2 μ s PREEMPTION

DRAM Queue: [MCU Safety ★ (QoS=15)] → Instant Priority Bypass
Preempts background video DMA in < 2 microseconds

- **Hardware QoS Priority Tags:** MCU Mailbox (QoS=15), NPU (QoS=10), Camera (QoS=4).
- **Zero Copy Ring Buffer:** Eliminates 3x user/kernel memory copying.
- **Deterministic Metrology:** Tail latency bounded at **P99 = 24.2 ms << τ_{world}** .